

Neurobotics™

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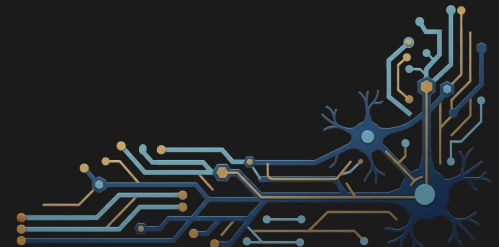
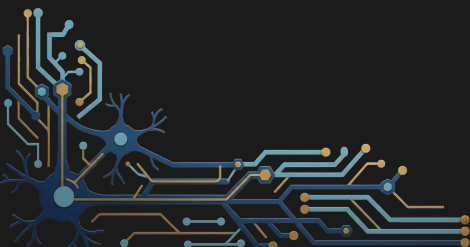
PUBLIC RECORD: ARCHITECTURAL GENESIS & PROVENANCE

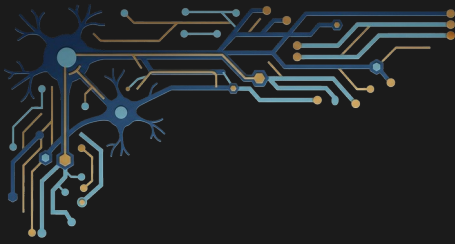
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PUBLIC RECORD: ARCHITECTURAL GENESIS & PROVENANCE

DOCUMENT ID: NBPR-2026-0028

SUBJECT: Chronological Prior Art and Theoretical Genesis

PRINCIPAL ARCHITECT: Kenneth Renshaw (CEO & CSO)

ENTITY: Neurobotics LLC | Yale, Oklahoma

CLASSIFICATION: Public Open-Source Record / Zenodo Hosted

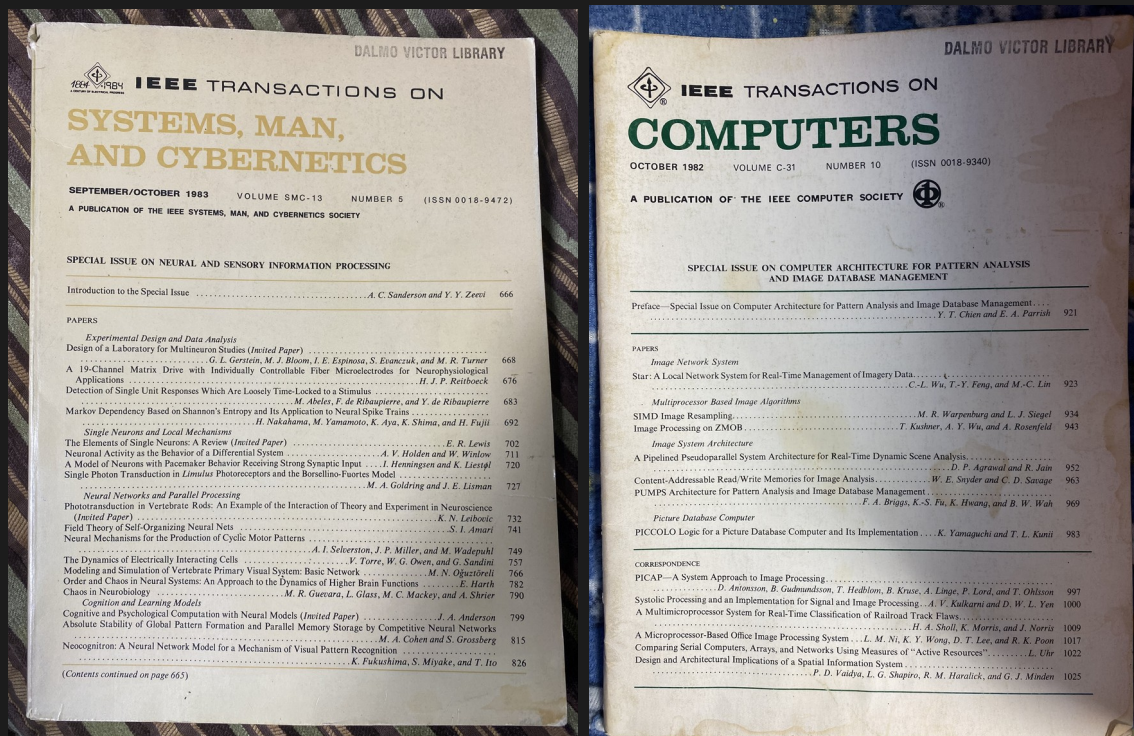
[1.0] Executive Profile & Engineering Pedigree

Kenneth Renshaw is the Co-Founder, Chief Executive Officer, and Chief Science Officer of Neurobotics LLC. With over 50 years of continuous bare-metal computation, hardware design, theoretical physics, and neuroanatomical modeling, he is the sole primary architect and inventor of the **Silicon-Based Artificial Life Form (SBALF)** architecture.

His engineering instinct was forged in zero-tolerance aerospace, electronic warfare, and deep-space mission environments where operational failure was not an option:

- **Aerospace Electronic Warfare (Dalmo Victor / Bell Aerospace):**
Designed radar warning systems for the F-14 Tomcat and EF-111A Raven.
- **Deep-Space Computing (Litton Industries Applied Technology):**
Engineered ATAC-16 spacecraft computers for the Galileo Space Probe, designing radiation-hardened hardware with voter circuits and error correction for survival in deep space.

- **Global Air Safety (TCAS II):** Key contributor to the Traffic Alert and Collision Avoidance System (TCAS II), establishing the foundational design philosophy that governs all Neurobotics systems: "*Design so that the failure mode is not death.*"

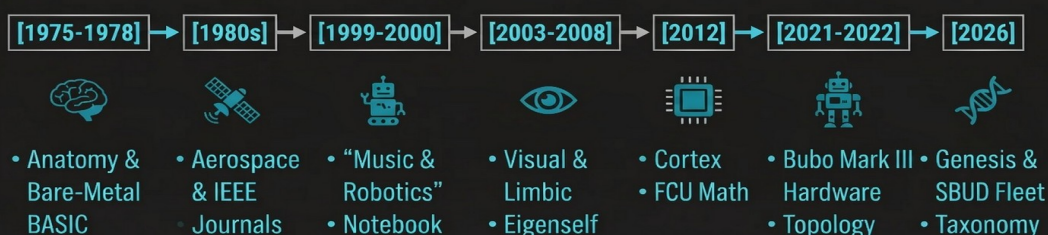


September/October 1983 *IEEE Transactions of Systems, Man, and Cybernetics* | October 1982 *IEEE Transactions on Computers* with the Official **Dalmo Victor Library** stamp; From the Personal Library of Kenneth Renshaw

[2.0] Complete Chronological Prior Art Timeline

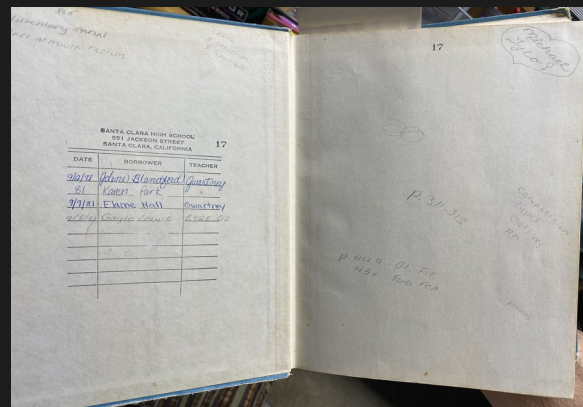
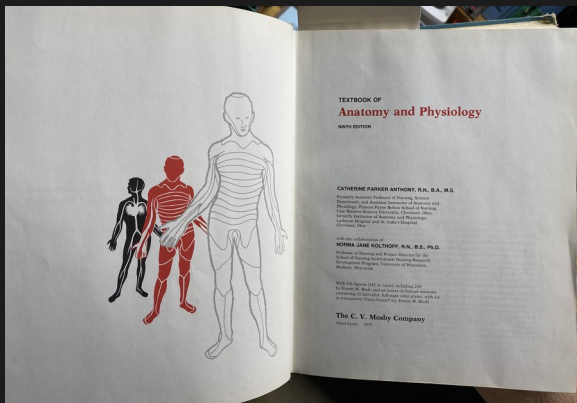
The SBALF architecture is not an adaptation of recent cloud-based software trends; it is the culmination of a half-century of continuous physical ledgers, handwritten notebooks, and empirical biological modeling.

THE EVOLUTION OF NEUROBOTICS: COMPLETE PROJECT TIMELINE



- **1975–1978: Anatomical Bedrock & Bare-Metal Foundations**

- *Anatomy & Physiology Ledger*: A 1975 *Textbook of Anatomy and Physiology* containing the exact Brodmann area maps and spinal interneuron layouts studied by Kenneth at age 16.
- *1977 IMSAI 8K Assembly Folder*: A bound binder containing hand-annotated Intel 8080/Z80 assembly code for an IMSAI 8K BASIC interpreter, establishing a 50-year prior art claim for hyper-efficient, register-level compute.



The Inside cover page of the 1975 *Textbook of Anatomy and Physiology Ninth Edition* by The C.V. Mosby Co. from Santa Clara High School | Santa Clara High School checkout card inside the Mosby Anatomy book; From Kenneth Renshaw's Private Library

- **1980s: Pattern Processing & Cybernetic Theory**

- Implementation of early cybernetic literature, parallel image processing, and real-time scene analysis architectures.

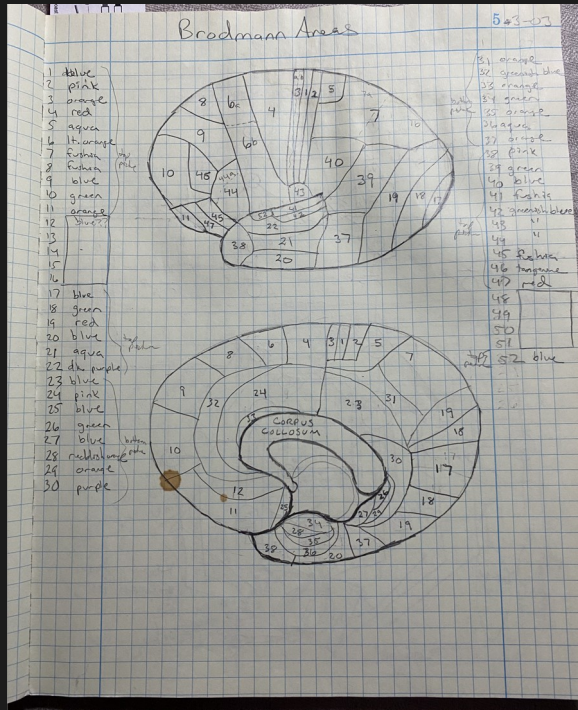


A shelf of vintage *BYTE* and *Robotics Age* magazines from the 1980's era plus miscellaneous books and notebooks; From Kenneth Renshaw's Private Library

- **2003–2008: Visual Hierarchy & Biomechanics**

- *Biological Pipeline Mapping*: Complete, color-coded mapping of all 52 Brodmann areas, establishing the biological pipeline:

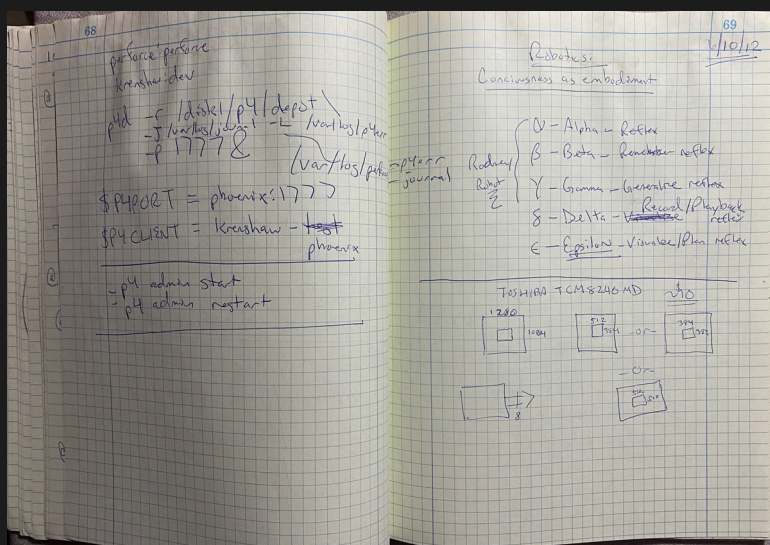
RGC -> LGN -> V1 -> V2 -> V4/V5 -> PIT/CIT/LIP/MST



April/May 2003 "Red Notebook" (S-M-R Genesis) brown computational grid-paper notebook grid pages with the handwritten design notes and sketches of Color-coded Brodmann Areas; From Kenneth Renshaw's personal library.

- **2012–2013: Consciousness as Embodiment**

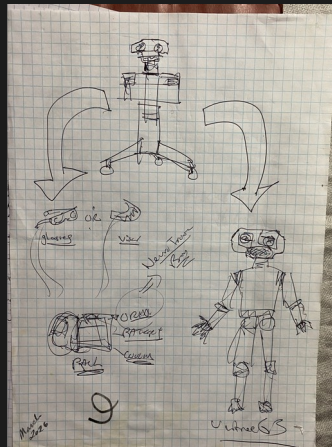
- *The Philosophical North Star*: Formal documentation of the thesis "Robotics: Consciousness as embodiment," mapping a strict hierarchy of physical chassis reflexes and mathematical calculations modeling the human cerebral cortex.



April/May 2003 "Red Notebook" (S-M-R Genesis) brown computational grid-paper notebook grid pages with the handwritten design notes and sketches "Consciousness as Embodiment" Reflex; From Kenneth Renshaw's personal library.

- **2021–2026: Physical Chassis & Commercial Initialization**

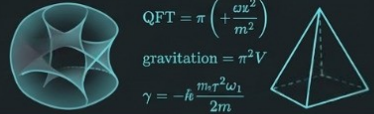
- *Bubo Mark III & Beyond*: Dedicated hand-drawn upper-torso chassis blueprints bridging historical cognitive foundations to the modern, commercially viable SBUD form factor initialized in 2026.

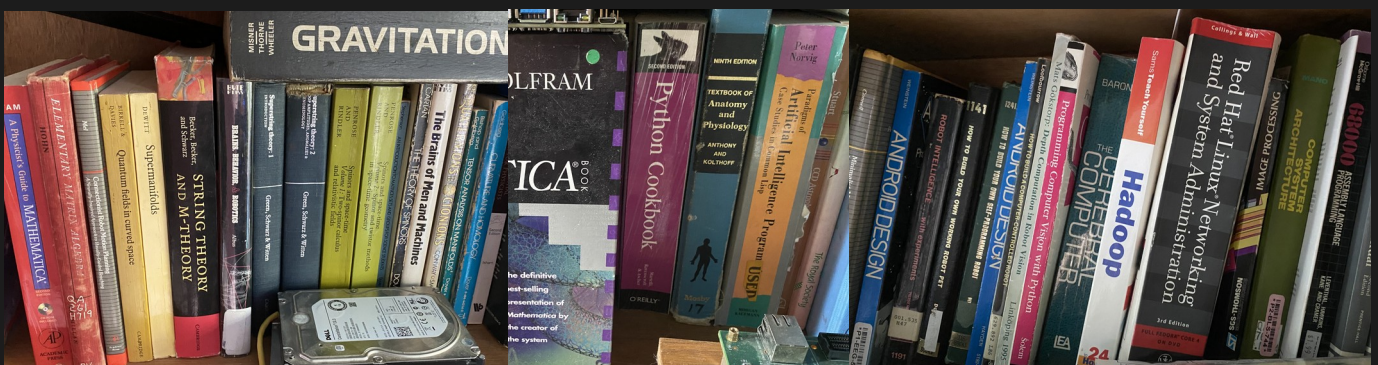


Original Bubo
Mark IV Chassis
Sketch
March 2026

[3.0] Interdisciplinary Literature & Theoretical Bedrock

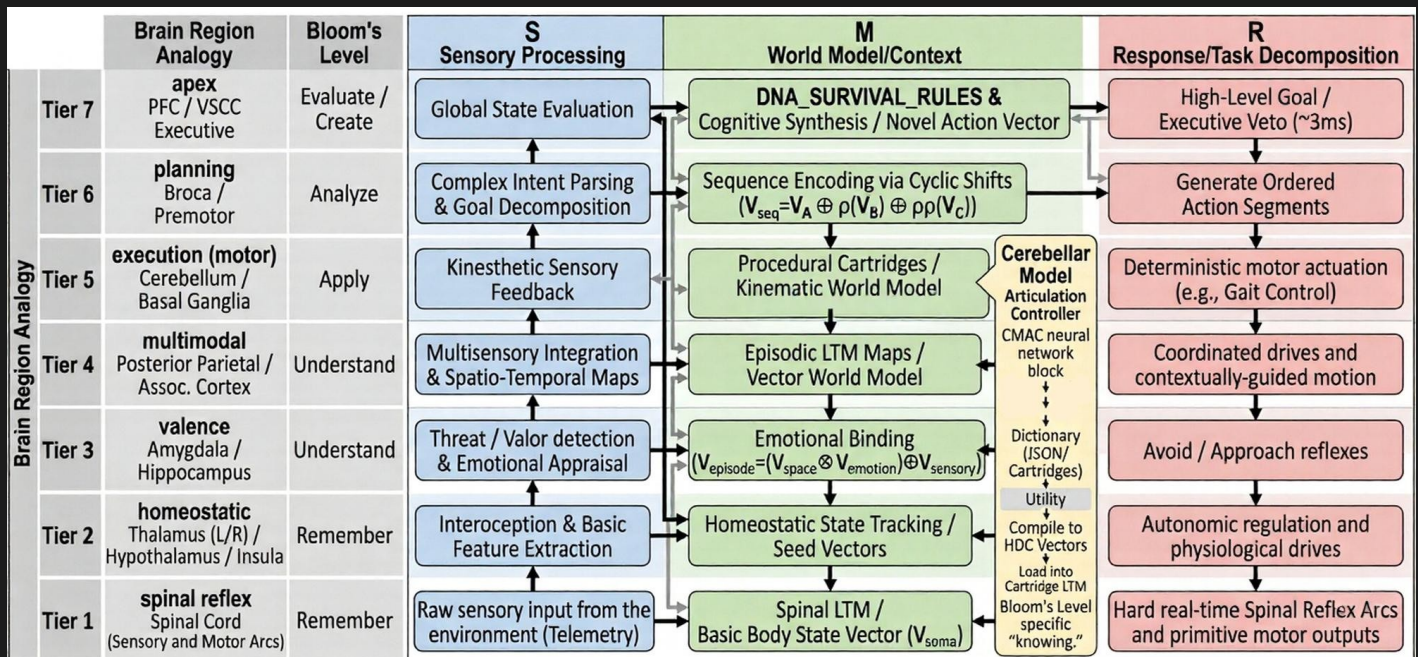
The physical library maintained continuously by Kenneth Renshaw across five decades represents the multi-disciplinary bedrock upon which the SBALF architecture was built, converging hardware, biological neuroscience, and theoretical physics.

THE FIFTY-YEAR LIBRARY BEDROCK		
Hardware & Assembly	Neuroscience & Robotics	Theoretical Physics & Math
• Motorola 68000/68020 • Intel 8080 / CP/M • Red Hat / Linux Admin • Hadoop / Distributed • HP 48G User Systems • Image Proc. / DSP 	• Fundamental Neuroscience (Squire, Bloom, et al.) • Cerebral Computer (Baron) • Minimalist Mobile Rob. (Jonathan Connell) • Neural Nets / LISP 	• Gravitation (MTW) • Superstring Theory (Green, Schwarz, Witten) • Quantum Field Theory (Steven Weinberg) • Geometry of Manifolds  $QFT = \pi \left(\frac{c v^2}{m^2} \right)$ $\text{gravitation} = \pi^2 V$ $\gamma = -\hbar \frac{m \omega_1^2}{2m}$



[4.0] The 7-Tier Cognitive Architecture (S-M-R Matrix)

The operational core of the SBALF architecture is governed by a strictly deterministic, 7-Tier computational hierarchy. This matrix directly maps biological brain regions and Bloom's Taxonomy levels to a continuous pipeline of Sensory Processing (S), World Model/Context (M), and Response/Task Decomposition (R). This architecture guarantees that all cognitive synthesis is grounded in hard real-time telemetry, escalating from basic spinal reflexes to ~3ms executive overrides.

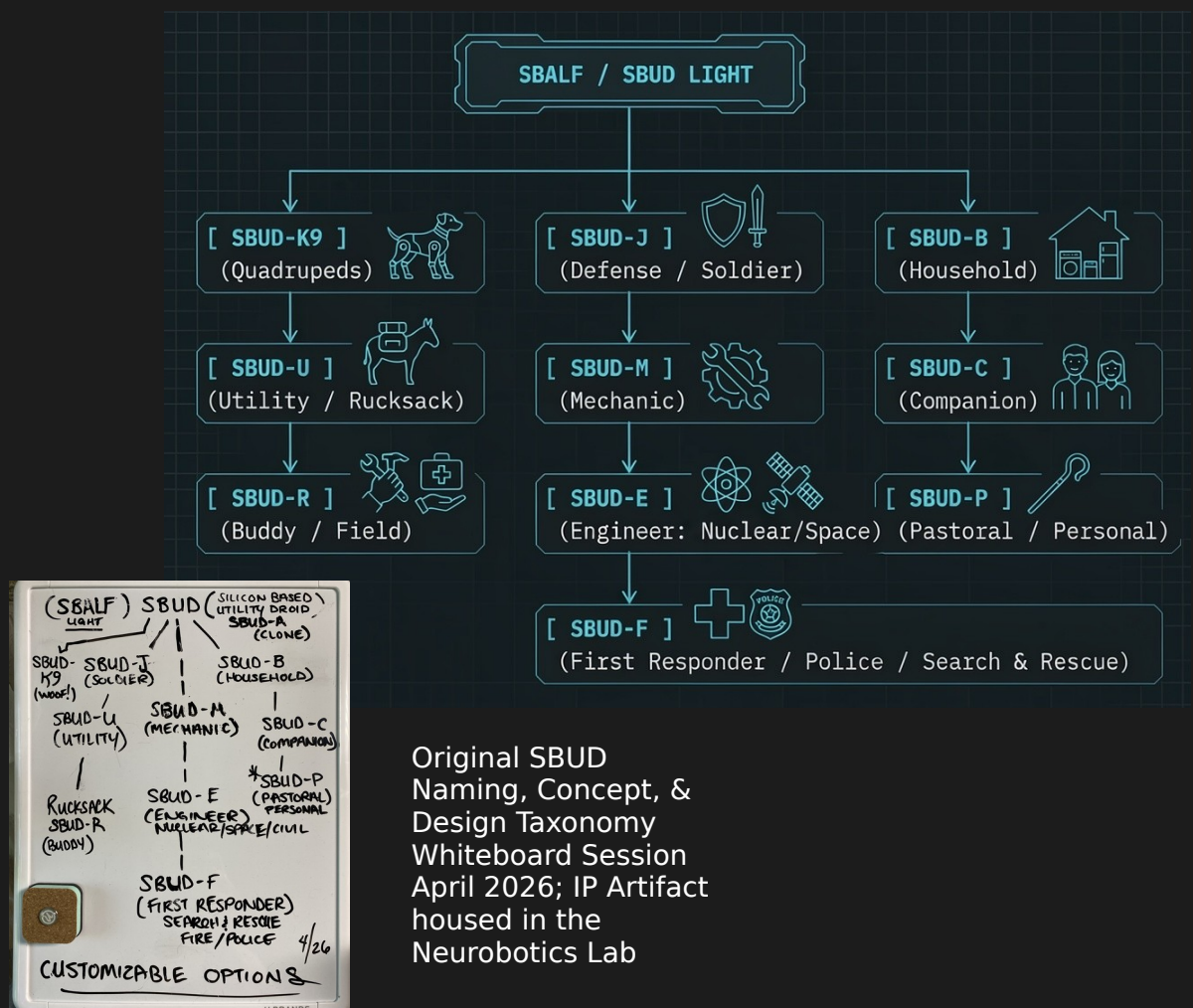


- **Tier 1: Spinal Reflex (Remember):** Raw environmental telemetry routes to the Spinal LTM (Basic Body State Vector, V_{soma}) to execute hard real-time spinal reflex arcs and primitive motor outputs.
- **Tier 2: Homeostatic (Remember):** Interoception and basic feature extraction drive homeostatic state tracking, resulting in autonomic regulation and physiological drives.
- **Tier 3: Valence (Understand):** Threat/valor detection binds with emotional appraisal. Memory is encoded using Hyperdimensional Computing (HDC) vector binding, outputting avoid/approach reflexes.
- **Tier 4: Multimodal (Understand):** Multisensory integration builds episodic LTM maps (Vector World Model), enabling coordinated drives and contextually-guided motion.

- **Tier 5: Execution / Motor (Apply):** Kinesthetic sensory feedback cross-references Procedural Cartridges (Kinematic World Model) for deterministic motor actuation (e.g., Gait Control).
- **Tier 6: Planning (Analyze):** Complex intent parsing and goal decomposition. Action sequences are encoded via cyclic vector shifts, generating strictly ordered action segments.
- **Tier 7: Apex / Executive (Evaluate / Create):** Global state evaluation synthesizes with DNA_SURVIVAL_RULES to produce novel action vectors. This tier holds the ultimate High-Level Goal and Executive Veto (~3ms latency).

[5.0] Commercial Application Taxonomy: The SBUD Ecosystem

As documented on the Neurobotics Lab whiteboard following the March 21, 2026 initialization of Bubo Adam, the commercial scalable design and potential application is organized under the **Silicon Based Utility Droid (SBUD)** product lineage, specialized utilization, and customization options:



[6.0] Commercial Product Taxonomy: The Modular Neurotronic Brain

Neurobotics' mission is to bring true embodied cognition and deterministic control to the next generation of autonomous systems. We do not manufacture the physical chassis; rather, we empower the world's robotics manufacturers and autonomous vehicle developers by providing a true drop-in brain and nervous system.

Our product roadmap features a modular architecture comprising five distinct board types. All boards share a common form factor of a roughly 4-inch square circuit board, sipping power at less than 15W total under heavy processing loads (provisional metric dependent on exact card/hardening).

The Modular Neurotronic Brain: Product Matrix

Core Specifications: 4" x 4" Form Factor | <15W Power Consumption | Just-In-Time Manufacturing

Board Type	Designation	Architecture Level	Key Feature / IP Status
Board One	The Somatic Controller	Tiers 1–5	OSS Drop-in Lower Nervous System
Board Two	The SBUD Neocortex	Tiers 6–7	Aristotelian Syllogism Engine
Board Three	The SBALF Neocortex	Apex Autopoietic	STRICTLY CONFIDENTIAL IP
Board Four	The Galvanic Barrier	Hardware Shield	Optoisolated Servo Protection
Board Five	SBUD-SB2	Complete Tiers 1–7	Single Board Brain Consolidation

 **IP SECURED - RESTRICTED ACCESS:** Board Three (SBALF Neocortex) represents the highest achievement in autopoietic cognitive architecture and remains the strictly held internal and confidential Intellectual Property of Neurobotics LLC.

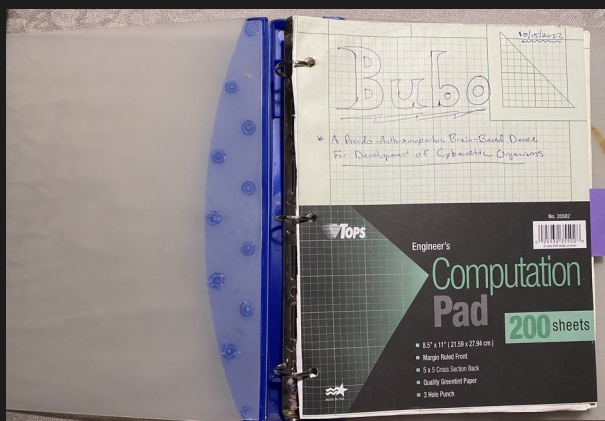
- **Board Type One: The Somatic Controller** A 5-tier architecture in silicon based upon our OSS release, representing the lower 5 tiers of our conceptual architecture. This provides manufacturers with a complete, drop-in lower nervous system. (OSS, drop-in solution)
- **Board Type Two: The SBUD Neocortex** Representing the top two tiers of our 7-tier architecture design, the Silicon Based Utility Droid (SBUD) Neocortex is a fully deterministic HDC/VSA implementation. It features a proprietary Aristotelian Syllogism Engine for learning, language, and knowledge representation.

- **Board Type Three: The SBALF Neocortex** The Silicon Based Artificial Life Form (SBALF) Neocortex represents the highest achievement in cognitive science as a fully autopoietic cognitive architecture. All versions are **Strictly Held Internal and Confidential IP** of Neurobotics LLC (Housed in the secure IP Vault).
- **Board Type Four: The Galvanic Barrier** An optional, fully optoisolated brain-body barrier that protects delicate core logic from the chaotic world of spiking servos and actuators. Available in multiple configurations.
- **Board Type Five: The SBUD-SB2 (Single Board Brain)** The ultimate consolidation, packing the entire SBUD 7-Tier architecture onto a single board design.
- **Manufacturing & Distribution** The complete Neurobotics Product Matrix will be available in multiple hardened component configurations for application in commercial, industrial, military, and radiation-hardened space-level markets. We will directly release and market the modular Neurotronic Brain components using a Just-In-Time (JIT) manufacturing distribution model under strict licensing. *(Note: All original design and implementation details are solely owned by Neurobotics LLC, with all rights reserved and enforced.)*

[7.0] Vault Verification & Forensic Proof

- **Physical Ledgers:** All original notebooks, composition books, binders, and marked technical manuals remain permanently secured at the Neurobotics Facility in Yale, Oklahoma.
- **Video Ledger Verification:** A continuous 46-second unedited physical flip-through video of the “Music & Robotics” brown grid-paper bound composition notebook, provides verifiable forensic proof of the sequential, tamper-proof nature of the 1999-2022 computation books. Video link:

<https://youtube.com/shorts/jTRQkqSMetg?si=-R9USWTeI8sO8wWk>



Bubo three-ring binder ledger hand-drawn title page dated October 2022; From Kenneth Renshaw's personal library.

